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KSAT Ranging Portal

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Michael.Johansen@ksat.no

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TDM-Angle

The pointing angles TDM file is used to provide the user with actual pointing angles commanded by the antenna controller. Pointing angles may be derived from a program track, autotrack, or during a scan.

Angle data files are indicated by the use of ANGLE as the prefix in the file name. The pointing angles file is not generated when readback of pointing angles from the antenna controller is not available.

An example TDM file for pointing angles is provided in [appendix A2](#).

Metadata Items

Metadata items relevant to this segment (in addition to the mandatory ones listed in [link](#)) are listed in the following table.

KEYWORD	VALUE
ANGLE_TYPE	
RECEIVE_BAND	S, X, Ka

The **ANGLE_TYPE** is set to indicate the type of 2-axis pedestals.

The **RECEIVE_BAND** indicates the frequency band used for the receive signal. This is relevant since it defines which autotracking receiver is in use (if autotracking is available at the site).

The TDM file format does not include provisions for indicating whether pointing angles are derived from program track, autotrack, or scanning patterns. However, this information can be critical for knowing whether the angle data can be used as part of flight dynamics analysis.

To provide this information to the user, an extension to the TDM format is provided using the COMMENT keyword in the metadata segment. For example:

```
COMMENT TRACKING_MODE = AUTO
COMMENT TRACKING_MODE = PROGRAM
```

All antenna angles provided in a given segment are derived using the same mode. If the mode of the antenna controller changes, the current segment is terminated, and a new segment is started, with the new mode indicated in the metadata comments. If tracking mode is added to a future TDM format, the TDM generator will be updated to incorporate tracking mode key/value entries into the metadata section.

Data Items

The data items for the angle data file are shown in the table below. The angle entries are mapped to one of the two axis angles as shown for the supported axis configurations.

KEYWORD	UNITS	RESOLUTION	AZEL	XEYN	XSYE
ANGLE_1	degrees	1e-6	azimuth	X-axis	X-axis
ANGLE_2	degrees	1e-6	elevation	Y-axis	Y-axis

-  Right-ascension/declination (RADEC) pedestals are not supported by this specification.
-  Pedestals with pre-positioned tilt axis are provisionally supported, though the antenna reference position will change on each pass. The updated reference position is provided in the TDM header.
-  Pedestals with dynamic 3-axis controls are not supported by TDM version 2.0.

Multi-Segments

The angle TDM files may contain one or more segments. The angles in each segment are generated by the antenna controller using the same mode (e.g. autotrack or program track). If the controller changes modes (starts autotracking, or drops out to program track), the current segment is terminated and a new segment is started.